

Beam Clamp

Technical Datasheet - Mechanical Theatre Equipment Compliance Portal

Generic editable template for inspection, maintenance and site records

Purpose

This datasheet provides a generic technical reference for rated beam clamps used to create attachment points from suitable structural beams or steel sections. It should be completed with the actual manufacturer data, certificates, installation information and inspection records for the specific item in use.

1. Equipment summary

Item	Information / site entry
Equipment type	Beam Clamp
Typical use	Rated attachment point for lifting, suspension or support from a suitable beam or steel section.
Manufacturer / model	To be completed from manufacturer marking or certificate.
Serial / ID / asset number	To be completed by site or inspection record.
Rated capacity	WLL/SWL to be confirmed from marking, certificate or manufacturer data.
Beam compatibility	Confirm beam flange width, flange thickness, profile and material are suitable for the clamp.
Load direction	Confirm approved load direction. Avoid side loading unless specifically allowed by the manufacturer.
Inspection interval	To be set by competent person / written scheme / site inspection regime.

2. Description and intended use

- A beam clamp is used to create a rated attachment point from a suitable structural beam or steel section.
- The clamp must be compatible with the beam flange size, flange thickness, beam profile, load direction and intended use.

- The beam, clamp, fixing method, suspended item and any attached lifting equipment should be assessed as one complete load path.
- Beam clamps should only be used within their stated WLL/SWL and in accordance with manufacturer instructions.
- The supporting structure must be suitable for the imposed load, including any dynamic effects and load combinations.

3. Regulatory and good-practice context

Reference	Applies where relevant	Practical meaning for beam clamps
LOLER 1998	Where the clamp is used as lifting equipment/accessory or as part of a lifting operation.	Selection, marking, strength, stability, planning, examination and defect reporting must be considered.
PUWER 1998	As work equipment used at work.	The clamp must be suitable, maintained, inspected where deterioration could cause risk, and used by competent persons.
BS EN 17206	Where used as part of entertainment machinery or a load-bearing entertainment system.	Compatibility with the complete load path and prevention of unsafe movement must be considered.
ABTT Technical Standards	As recognised theatre-industry good practice.	Supports competent installation, inspection, record keeping and venue-specific assessment.

Important

This is not a substitute for the manufacturer manual, structural assessment, certificate of conformity, thorough examination report or written scheme of examination. Where rating, compatibility or structural suitability is uncertain, the clamp should not be used until confirmed by a competent person.

4. Selection and installation checks

- Confirm the clamp type is suitable for the intended operation and environment.
- Confirm WLL/SWL, identification and traceability markings are present and legible.
- Confirm the beam is suitable for the load, load direction and point of application.
- Confirm flange width and thickness are within the manufacturer's stated adjustment range.
- Confirm the clamp is seated squarely and fully engaged on the beam flange.
- Do not use on damaged, excessively corroded, unsuitable, tapered or non-standard beams unless approved by a competent person.
- Avoid shock loading, side loading, eccentric loading or use outside the manufacturer's stated configuration.
- Use compatible shackles, eyes, hooks or suspension hardware so that the load is applied correctly.

5. Inspection checklist

No.	Check area	Inspection guidance
1	Identification and rating	Check WLL/SWL, manufacturer, size and traceability markings. Do not use if the rating is unknown or illegible.
2	Clamp body and jaws	Inspect for cracks, distortion, bends, corrosion, impact damage, wear, gouges or signs of overload.
3	Spindle / screw / bolts	Check thread condition, smooth operation, full engagement, locking parts and absence of damage or seizure.
4	Suspension eye / attachment point	Inspect the eye, lug or attachment point for wear, elongation, distortion, cracking or incorrect orientation.
5	Beam compatibility	Confirm the beam flange size, thickness, profile and condition are suitable for the clamp.
6	Seating and alignment	Check that the clamp sits squarely, grips correctly and cannot slip, twist or walk along the beam under load.
7	Load direction	Confirm the load is applied in the approved direction and does not introduce unapproved side loading or bending.
8	Supporting structure	Check the parent beam and support structure for corrosion, cracking, movement, deformation or other visible distress.
9	Associated accessories	Inspect shackles, slings, hooks, hoists or other connected items for compatibility and condition.
10	Records and defects	Confirm inspection records are current and that defects have been reported, quarantined and actioned.

6. Removal from service triggers

- Unknown, missing or illegible WLL/SWL or identification marking.
- Cracks, distortion, bent body, damaged jaws, excessive wear, corrosion or signs of overloading.
- Damaged or seized threads, missing locking components, poor engagement or incorrect assembly.
- Evidence of slipping, movement, rotation, point loading or incorrect load direction.
- Use on unsuitable beam, doubtful structure, unknown rating or unsupported installation method.
- Unauthorised drilling, welding, grinding, heating, repair or modification.
- Missing inspection records where required or any uncertainty over condition or compatibility.

7. Site-specific completion record

Item	Information / site entry
Venue / site	
Location / beam reference	
Asset ID	
Manufacturer / model	
WLL/SWL	
Beam size / flange range confirmed	
Approved load direction	
Certificate / report reference	
Inspection frequency	
Competent person / inspector	
Notes / restrictions	

8. Portal notes

- Suggested portal file name: documents/beam_clamp_technical_datasheet.pdf
- Suggested portal title: Beam Clamp Technical Datasheet
- This document is intended to sit under Technical data / extra information for the Beam Clamp equipment record.